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NRC7394 Evaluation Kit User Guide

(Raspberry Pi setup)

Ultra-low power & Long-range Wi-Fi

Ver 1.3
Sep. 29, 2026

NEWRACOM, Inc.

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NRC7394 Evaluation Kit User Guide (Raspberry Pi setup)
Ultra-low power & Long-range Wi-Fi

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1 Overview

The NRC7394 Evaluation Kit (EVK) uses a Raspberry Pi 4 model B board as a Linux host. However, the Raspberry Pi 3 model B/B+ board can be also used.

- Raspberry Pi 3 Model B : https://www.raspberrypi.org/products/raspberry-pi-3-model-b/



Figure 1.1 Raspberry Pi 3 Model B Board

- Raspberry Pi 3 Model B+ : https://www.raspberrypi.org/products/raspberry-pi-3-model-b-plus/



Figure 1.2 Raspberry Pi 3 Model B+ Board

- Raspberry Pi 4 Model B : https://www.raspberrypi.org/products/raspberry-pi-4-model-b/

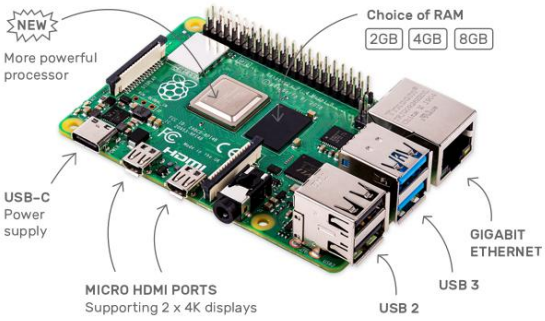


Figure 1.3 Raspberry Pi 4 Model B Board

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Figure 1.4 shows the connection between the NRC7394 evaluation board and the Raspberry Pi 4 model B board.

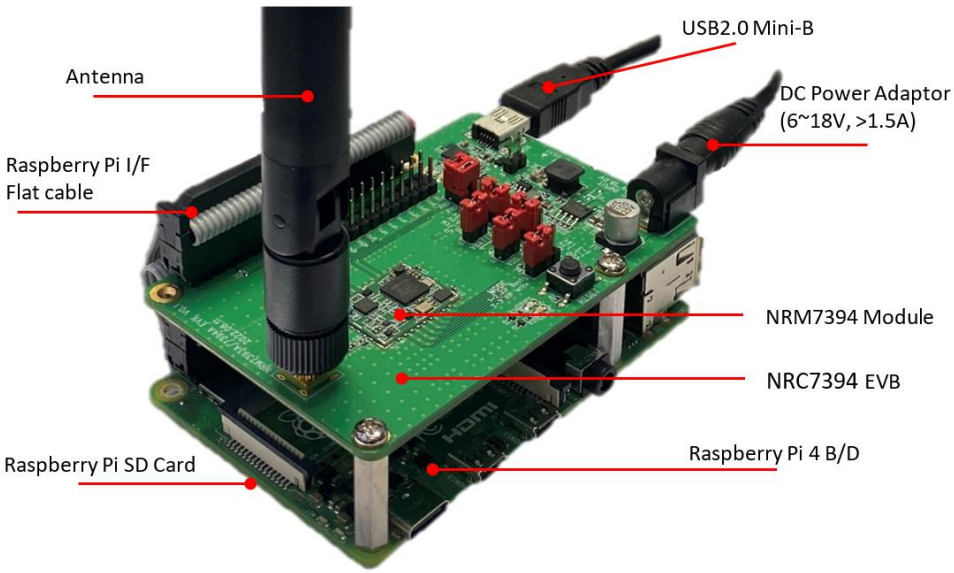


Figure 1.4 NRC7394 Evaluation Kit with Raspberry Pi 4 model B

Raspberry Pi OS is the recommended operating system for normal use on Raspberry Pi boards. And for the NRC7394 Evaluation Kit, it requires some additional configuration.

- Enable SSH or VNC (optional)
- Enable SPI and Serial Port
- Disable Broadcom Wi-Fi and Bluetooth
- Disable User mode SPI device driver
- Upgrade the Linux kernel (optional)
- Install the Linux kernel headers
- Install the required packages, etc.

This document describes how to configure a Raspberry Pi image for the NRC7394 Evaluation Kit, which includes the NRC7394 Software Package.

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2 Installing Raspberry Pi OS

Raspberry Pi recommends the use of Raspberry Pi Imager to install an operating system on the SD card. The Raspberry Pi OS can be quickly and easily installed on a microSD card using Raspberry Pi Imager.

Prepare the following:

- A computer with a microSD card reader
- A microSD card of at least 8 GB in size

2.1 Download and install Raspberry Pi Imager

Download the Raspberry Pi Imager for Windows at https://www.raspberrypi.org/software/.

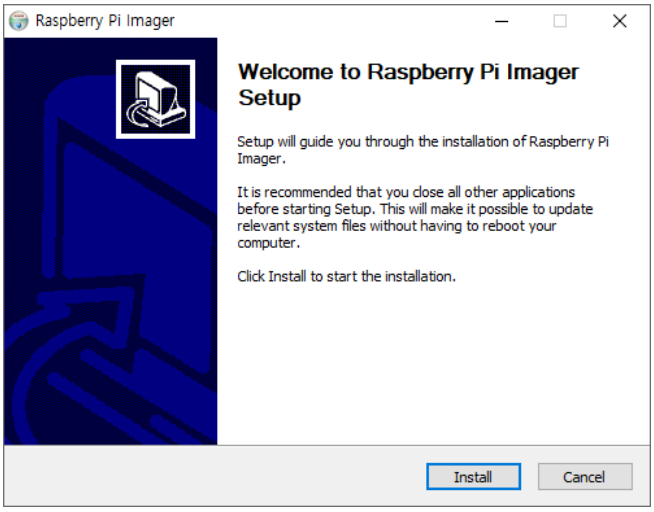


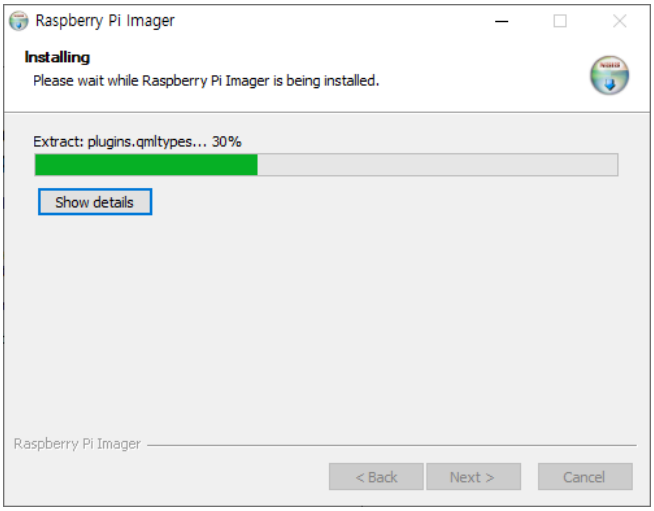
Figure 2.1 Raspberry Pi Imager Download Page

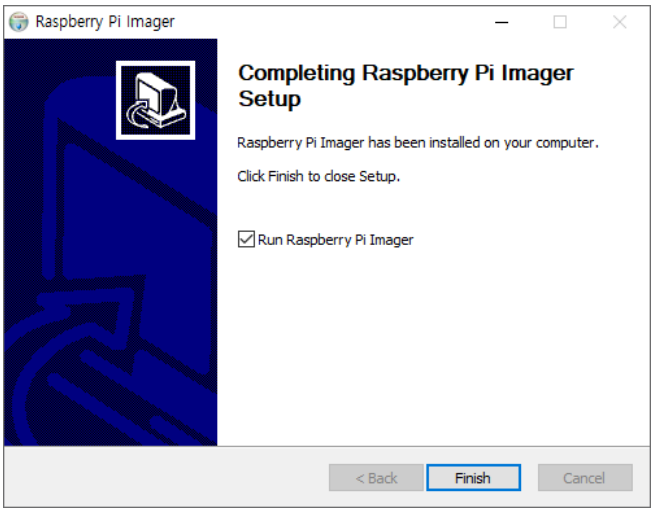
Raspberry Pi Imager for Windows : https://downloads.raspberrypi.org/imager/imager_latest.exe

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When the download finishes, click it to launch the installer and follow as below.







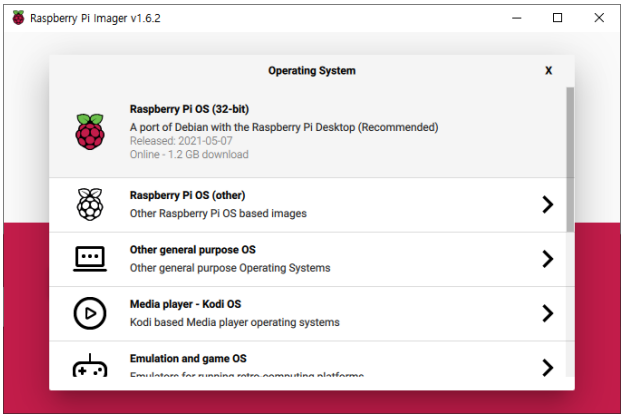
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2.2 Write Raspberry Pi OS image on microSD card

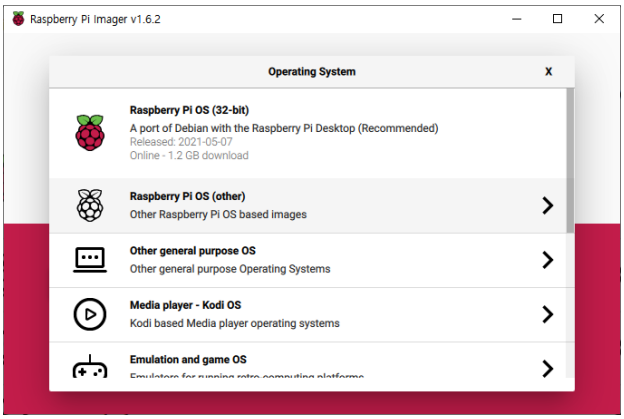
After inserting the microSD card into a computer with a microSD card reader, launch the Raspberry Pi Imager. And click the "CHOOSE OS" button to choose the operating system.



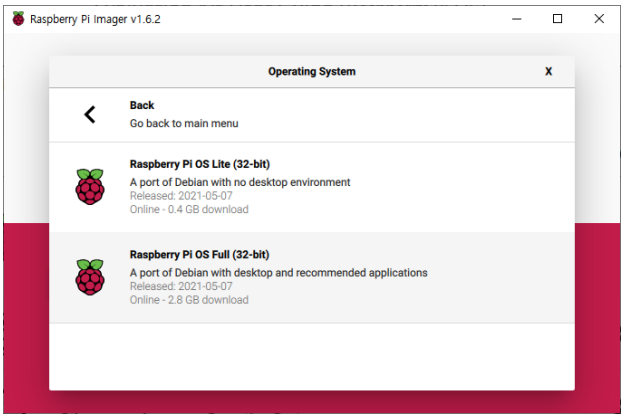
Select the “Raspberry Pi OS (32-bit)” menu to install the latest Raspberry Pi OS. The release date is May 7, 2021 when this document is written.



Or select the "Raspberry Pi OS (other)" menu to install the Raspberry Pi OS of other size.



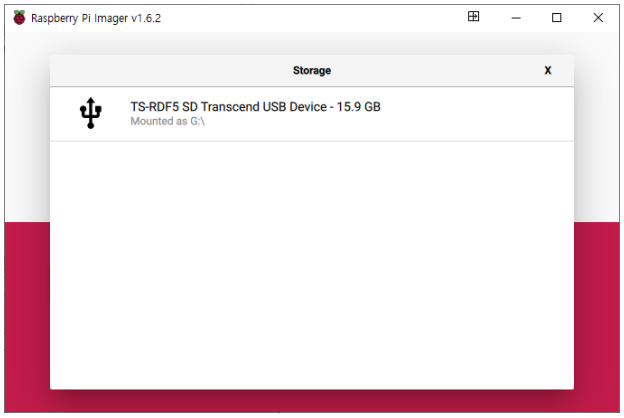
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Click the “CHOOSE STORAGE” button to choose the SD card.

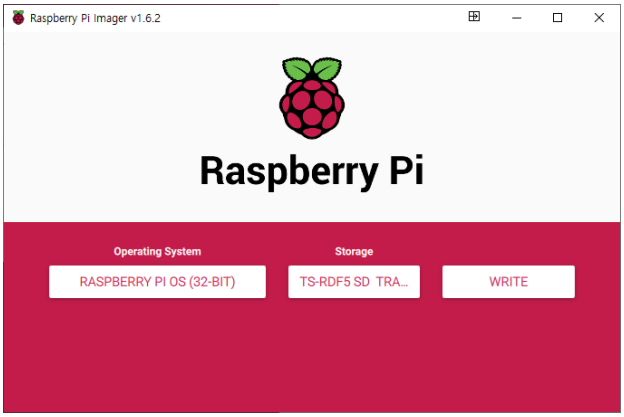


Select the microSD card to write the Raspberry Pi OS image.



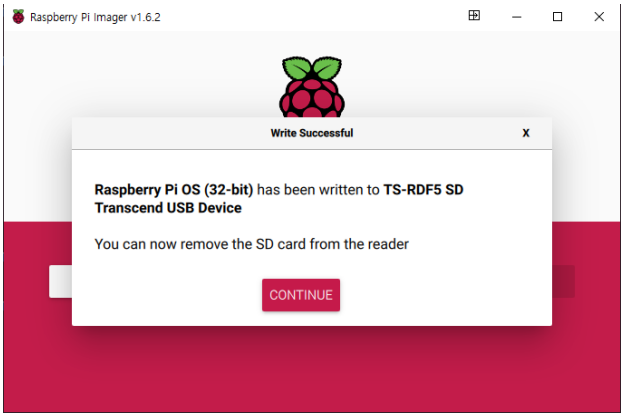
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Click the “WRITE” button and wait for the Raspberry Pi Imager to finish writing.





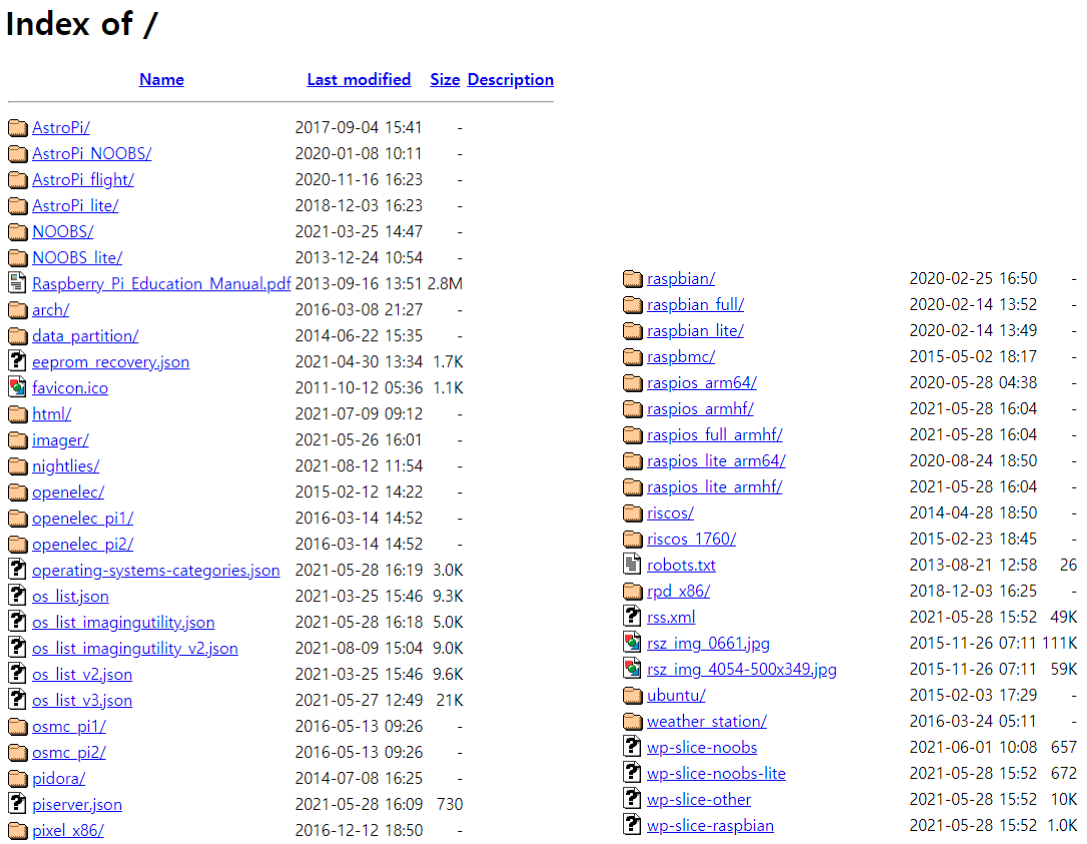
Click the “CONTINUE” button to eject the microSD card after the following message.



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NOTE:

Previous versions of the Raspberry Pi OS can be downloaded from http://downloads.raspberrypi.org/.

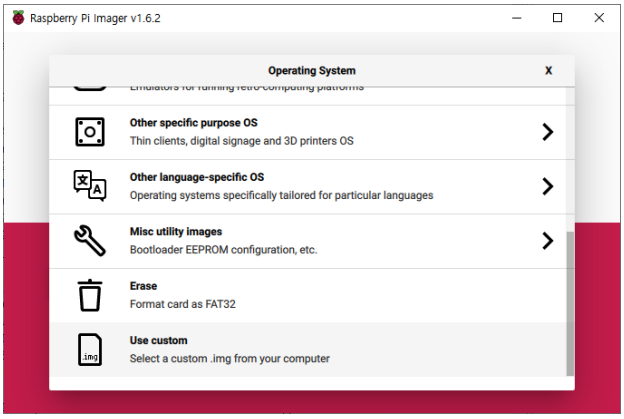


Index of /

Name	Last modified	Size	Description
AstroPi/	2017-09-04 15:41	-	
AstroPi_NOOBS/	2020-01-08 10:11	-	
AstroPi_flight/	2020-11-16 16:23	-	
AstroPi_lite/	2018-12-03 16:23	-	
NOOBS/	2021-03-25 14:47	-	
NOOBS_lite/	2013-12-24 10:54	-	
Raspberry_Pi_Education_Manual.pdf	2013-09-16 13:51	2.8M	
arch/	2016-03-08 21:27	-	
data_partition/	2014-06-22 15:35	-	
eeprom_recovery.json	2021-04-30 13:34	1.7K	
favicon.ico	2011-10-12 05:36	1.1K	
html/	2021-07-09 09:12	-	
imager/	2021-05-26 16:01	-	
nightlies/	2021-08-12 11:54	-	
openelec/	2015-02-12 14:22	-	
openelec_pi1/	2016-03-14 14:52	-	
openelec_pi2/	2016-03-14 14:52	-	
operating-systems-categories.json	2021-05-28 16:19	3.0K	
os_list.json	2021-03-25 15:46	9.3K	
os_list_imagingutility.json	2021-05-28 16:18	5.0K	
os_list_imagingutility_v2.json	2021-08-09 15:04	9.0K	
os_list_v2.json	2021-03-25 15:46	9.6K	
os_list_v3.json	2021-05-27 12:49	21K	
osmc_pi1/	2016-05-13 09:26	-	
osmc_pi2/	2016-05-13 09:26	-	
pidora/	2014-07-08 16:25	-	
piserver.json	2021-05-28 16:09	730	
pixel_x86/	2016-12-12 18:50	-	
raspbian/	2020-02-25 16:50	-	
raspbian_full/	2020-02-14 13:52	-	
raspbian_lite/	2020-02-14 13:49	-	
raspbmc/	2015-05-02 18:17	-	
raspios_arm64/	2020-05-28 04:38	-	
raspios_armhf/	2021-05-28 16:04	-	
raspios_full_armhf/	2021-05-28 16:04	-	
raspios_lite_arm64/	2020-08-24 18:50	-	
raspios_lite_armhf/	2021-05-28 16:04	-	
riscos/	2014-04-28 18:50	-	
riscos_1760/	2015-02-23 18:45	-	
robots.txt	2013-08-21 12:58	26	
rpd_x86/	2018-12-03 16:25	-	
rss.xml	2021-05-28 15:52	49K	
rsz_img_0661.jpg	2015-11-26 07:11	111K	
rsz_img_4054-500x349.jpg	2015-11-26 07:11	59K	
ubuntu/	2015-02-03 17:29	-	
weather_station/	2016-03-24 05:11	-	
wp-slice-noobs	2021-06-01 10:08	657	
wp-slice-noobs-lite	2021-05-28 15:52	672	
wp-slice-other	2021-05-28 15:52	10K	
wp-slice-raspbian	2021-05-28 15:52	1.0K	

Figure 2.2 Raspberry Pi Download Page

The downloaded Raspberry Pi OS image can also be selected in Raspberry Pi Imager and written to the SD card. Click the “Use custom” menu in the “Operating System” window.



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The release date of Raspberry Pi OS that supports Raspberry Pi 3 Model B/B+ and Raspberry Pi 4 Model B can be found at https://en.wikipedia.org/wiki/Raspberry_Pi_OS.

Raspberry Pi Model	Raspberry Pi OS Release Date
Raspberry Pi 3 Model B	2016-02-09 ~ latest
Raspberry Pi 3 Model B+	2018-04-18 ~ latest
Raspberry Pi 4 Model B	2019-06-24 ~ latest

Table 2.1 Supported Raspberry Pi OS Release Dates

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Release date	Debian version	Linux Kernel	GCC	APT	X Server	Pi 1/1+	Pi 2	Pi 3	Pi Zero W	Pi 3+	Pi 4	Pi Zero 2 W
2013-09-27	7 (Wheezy)	3.6	4.7.2	0.9.7	7.7	✓	✘	✘	✘	✘	✘	✘
2013-10-07						✓	✘	✘	✘	✘	✘	✘
2013-12-24		3.10				✓	✘	✘	✘	✘	✘	✘
2014-01-09						✓	✘	✘	✘	✘	✘	✘
2014-06-22		3.12				✓	✘	✘	✘	✘	✘	✘
2014-07-08						✓	✘	✘	✘	✘	✘	✘
2014-09-12						✓	✘	✘	✘	✘	✘	✘
2014-10-08						✓	✘	✘	✘	✘	✘	✘
2014-12-25						✓	✘	✘	✘	✘	✘	✘
2015-02-02		3.18				✓	✓	✘	✘	✘	✘	✘
2015-02-17						✓	✓	✘	✘	✘	✘	✘
2015-02-18						✓	✓	✘	✘	✘	✘	✘
2015-05-07						✓	✓	✘	✘	✘	✘	✘
2015-05-12						✓	✓	✘	✘	✘	✘	✘
2015-09-28	8 (Jessie)	4.1	4.9	1.0.9.8.1		✓	✓	✘	✘	✘	✘	✘
2015-11-24						✓	✓	✘	✘	✘	✘	✘
2016-02-08						✓	✓	✘	✘	✘	✘	✘
2016-02-09						✓	✓	✘	✘	✘	✘	✘
2016-02-29						✓	✓	✓	✘	✘	✘	✘
2016-03-18						✓	✓	✓	✘	✘	✘	✘
2016-05-13		4.4				✓	✓	✓	✘	✘	✘	✘
2016-05-31						✓	✓	✓	✘	✘	✘	✘
2016-09-28						✓	✓	✓	✘	✘	✘	✘
2016-11-29						✓	✓	✓	✘	✘	✘	✘
2017-02-27		4.9				✓	✓	✓	✘	✘	✘	✘
2017-03-03						✓	✓	✓	✓	✘	✘	✘
2017-04-10						✓	✓	✓	✓	✘	✘	✘
2017-06-23						✓	✓	✓	✓	✘	✘	✘
2017-07-05						✓	✓	✓	✓	✘	✘	✘
2017-08-17	9 (Stretch)		6.3	1.4.6		✓	✓	✓	✓	✘	✘	✘
2017-09-08						✓	✓	✓	✓	✘	✘	✘
2017-11-29						✓	✓	✓	✓	✘	✘	✘
2018-03-13						✓	✓	✓	✓	✘	✘	✘
2018-04-18		4.14		1.4.8		✓	✓	✓	✓	✓	✘	✘
2018-06-29						✓	✓	✓	✓	✓	✘	✘
2018-10-09						✓	✓	✓	✓	✓	✘	✘
2018-11-13						✓	✓	✓	✓	✓	✘	✘
2019-04-08				1.4.9		✓	✓	✓	✓	✓	✘	✘
2019-06-24	10 (Buster)	4.19	8.3	1.8.2		✓	✓	✓	✓	✓	✓	✘
2019-07-10						✓	✓	✓	✓	✓	✓	✘
2019-09-30						✓	✓	✓	✓	✓	✓	✘
2020-02-07						✓	✓	✓	✓	✓	✓	✘
2020-02-14						✓	✓	✓	✓	✓	✓	✘
2020-05-27						✓	✓	✓	✓	✓	✓	✘
2020-08-20		5.4.51				✓	✓	✓	✓	✓	✓	✘
2020-12-02		5.4.79		1.8.2.1		✓	✓	✓	✓	✓	✓	✘
2021-01-11		5.4.83		1.8.2.2		✓	✓	✓	✓	✓	✓	✘
2021-03-04		5.10.17				✓	✓	✓	✓	✓	✓	✘
2021-05-07				1.8.2.3		✓	✓	✓	✓	✓	✓	✓
2021-10-30		5.10.63				✓	✓	✓	✓	✓	✓	✓
2021-12-03	11 (Bullseye)		10.2.1	2.2.4	1.20.11	✓	✓	✓	✓	✓	✓	✓
2022-01-28		5.10.92				✓	✓	✓	✓	✓	✓	✓
2022-03-08		5.10.103				✓	✓	✓	✓	✓	✓	✓
2022-04-04		5.15.30				✓	✓	✓	✓	✓	✓	✓
2022-09-06		5.15.61				✓	✓	✓	✓	✓	✓	✓
2022-09-22						✓	✓	✓	✓	✓	✓	✓
2023-02-21		5.15.84				✓	✓	✓	✓	✓	✓	✓
2023-05-03		6.1.21				✓	✓	✓	✓	✓	✓	✓

Figure 2.3 Raspberry Pi OS Release History

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3 Setting up Raspberry Pi

To set up the Raspberry Pi, prepare the following and connect it to the Raspberry Pi board as shown in Figure 3.1.

- A 5V power adapter (optional)
- Keyboard and mouse
- TV or PC monitor with HDMI port
- HDMI cable
  - Raspberry Pi 3 : standard HDMI
  - Raspberry Pi 4 : Micro HDMI
- Ethernet cable with internet access

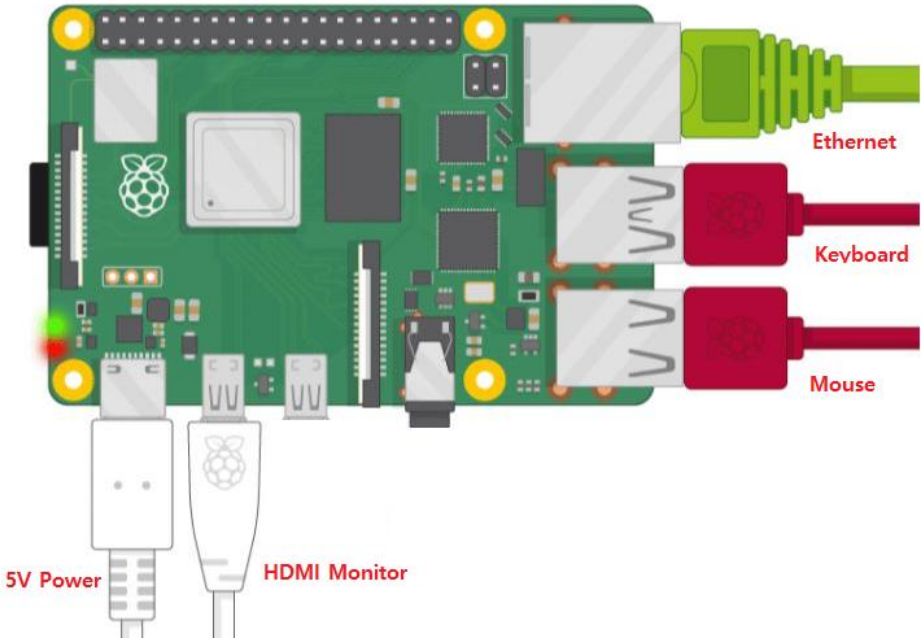


Figure 3.1 Raspberry Pi Model 4 B Cable Connection

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3.1 Enable the required interfaces

Launch the Raspberry Pi configuration tool.

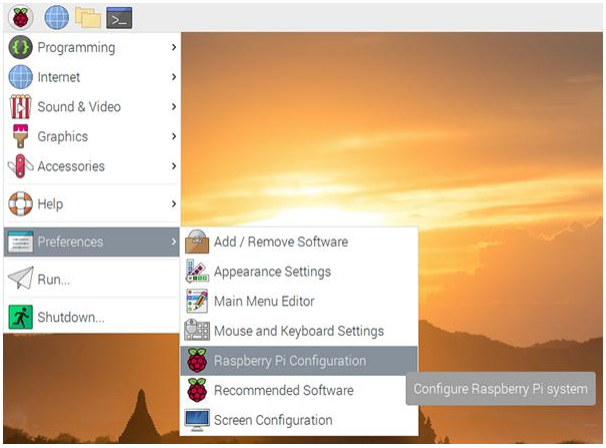
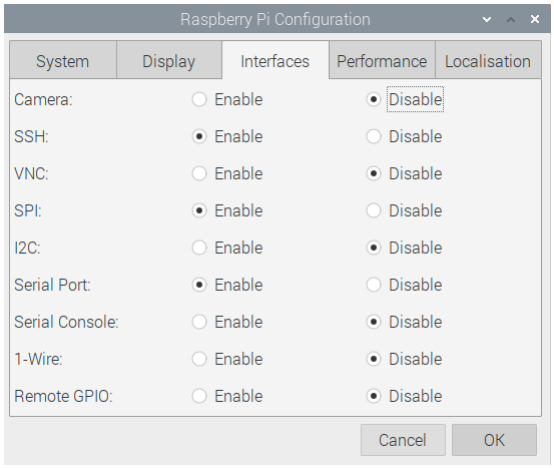


Figure 3.2 Raspberry Pi Configuration Tool Menu

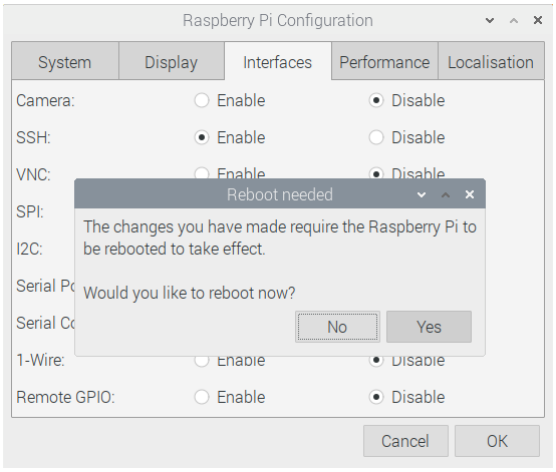
Enable the required interfaces in the “Interfaces” tap.

- Enable SSH or VNC for remote access (optional)
- Enable SPI and Serial Port to communicate with the NRC7394
  - The SPI is used in Linux host mode and standalone mode that supports AT Command
  - The Serial Port is used only in standalone mode that supports AT Command
- Disable serial console so that the console log is not output via the Serial Port



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After completing all configuration, reboot the system.



NOTE:

The Raspberry Pi configuration tool can also be run from the command line.

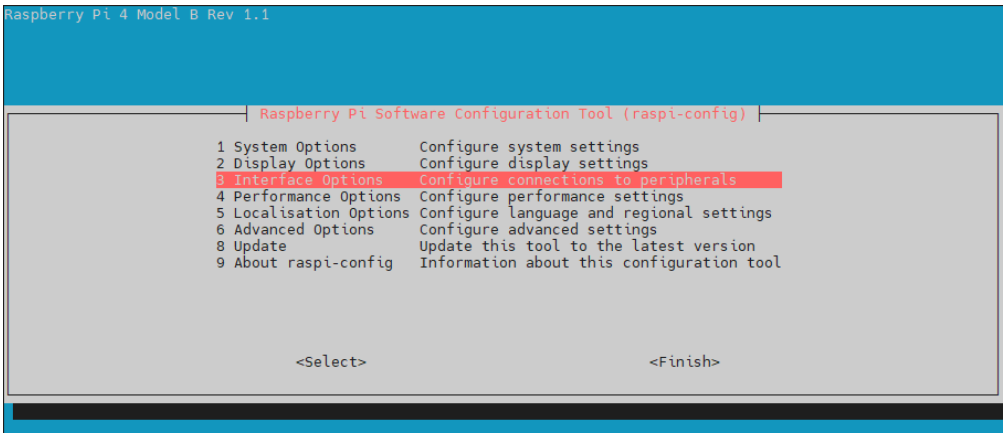
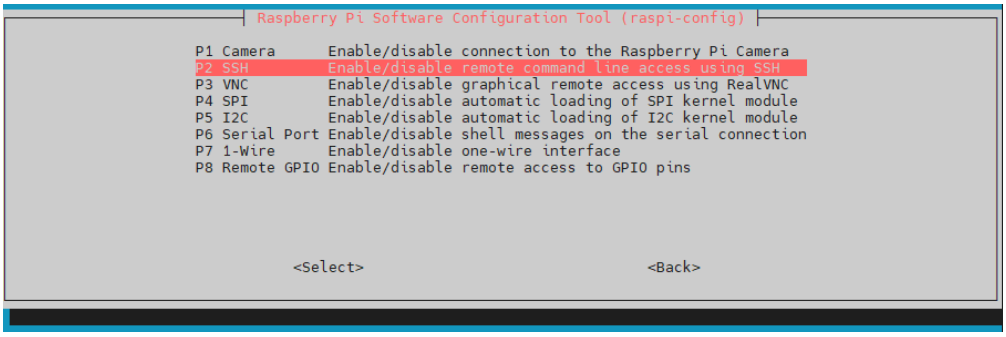
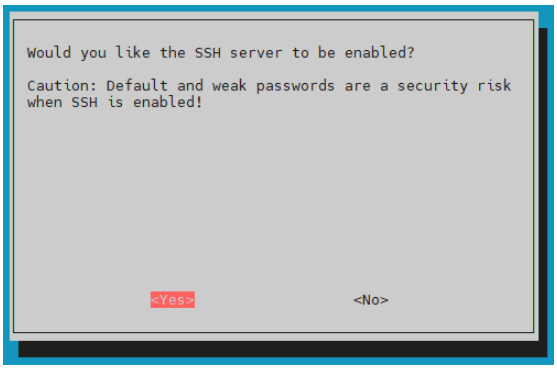


Figure 3.3 Raspberry Pi Configuration Tool (command line)

- Enable SSH

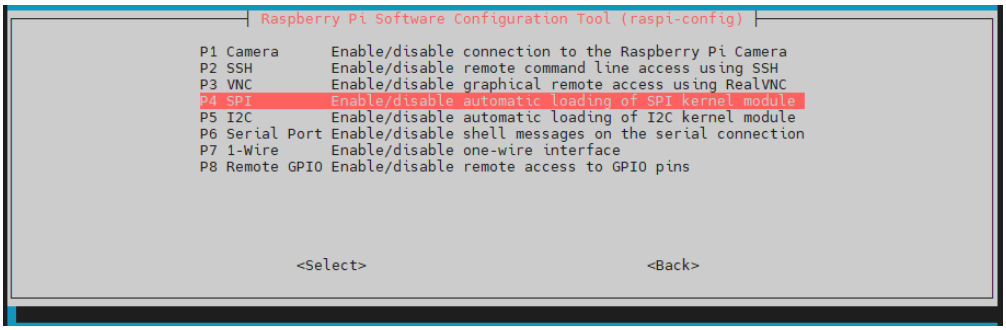


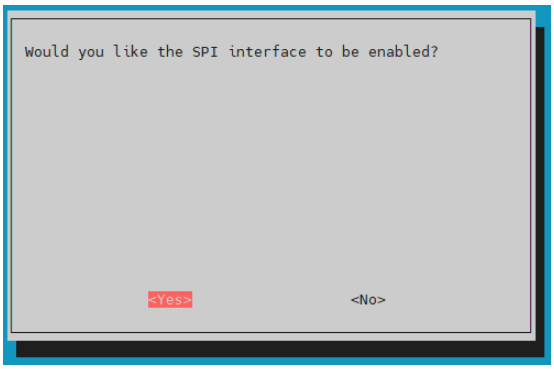
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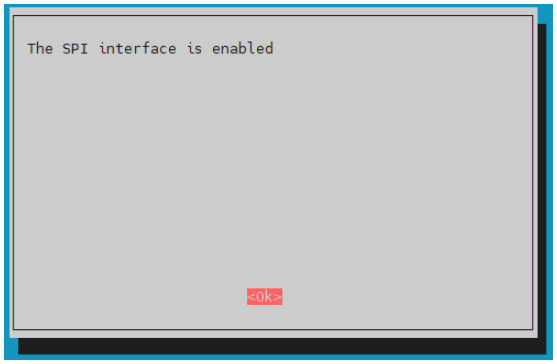




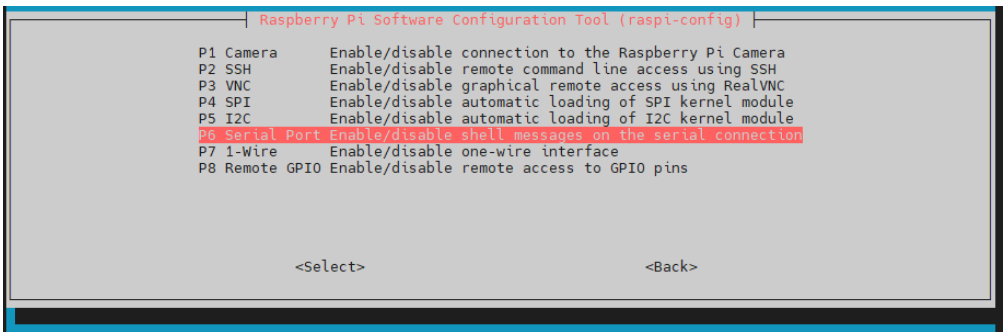
- Enable SPI



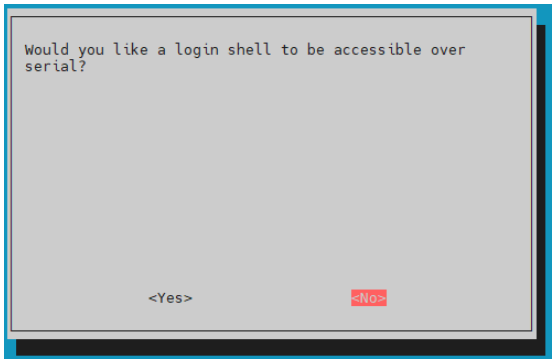


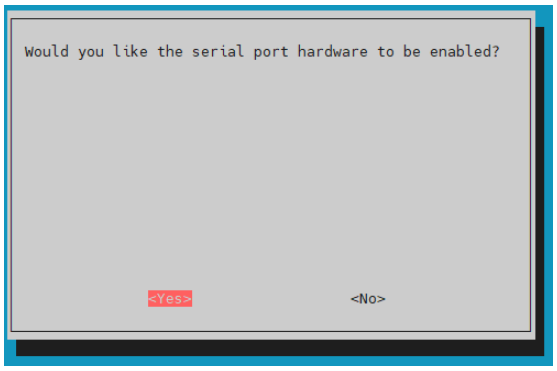


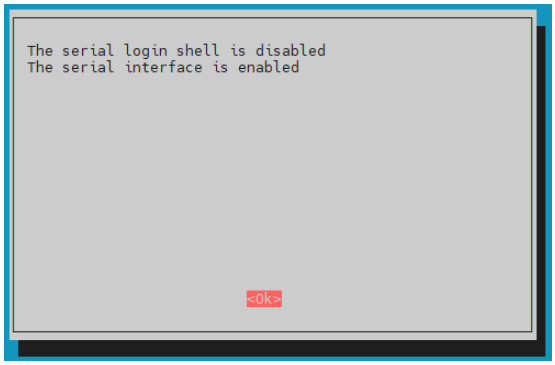
- Enable Serial Port



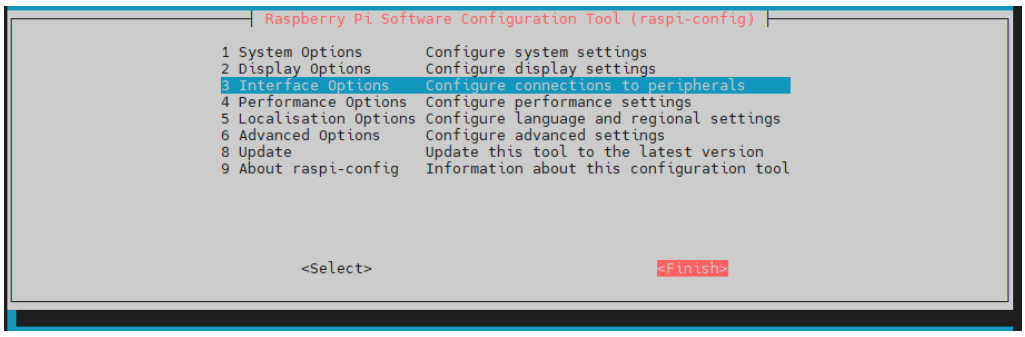
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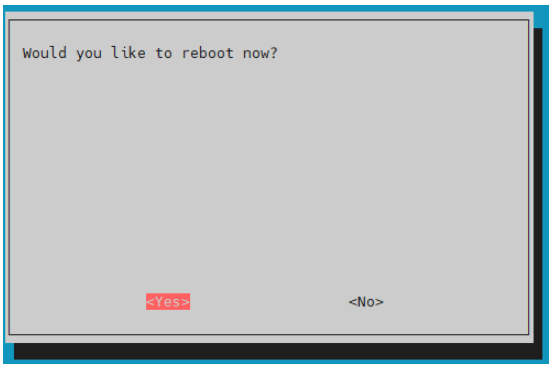






- Fininsh and reboot





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3.2 Install the required packages

Update the list of available packages.

```
$ sudo apt update
```

Upgrade installed packages, including the Linux kernel. If there is a newer version than the current kernel, it will be upgraded. Upgrading the package is optional.

```
$ sudo apt upgrade
```

Alternatively, only the kernel can be upgraded to the latest version by installing the raspberrypi-bootloader and raspberrypi-kernel packages.

```
$ uname -a
```

```
Linux raspberrypi 5.10.17-v7l+ #1414 SMP Fri Apr 30 13:20:47 BST 2021 armv7l GNU/Linux
```

```
$ ls -l /lib/modules
```

```
drwxr-xr-x 3 root root 4096 May  7 15:41 5.10.17+
drwxr-xr-x 3 root root 4096 May  7 15:41 5.10.17-v7+
drwxr-xr-x 3 root root 4096 May  7 15:41 5.10.17-v7l+
drwxr-xr-x 3 root root 4096 May  7 15:41 5.10.17-v8+
```

```
$ sudo apt install raspberrypi-bootloader raspberrypi-kernel
```

```
$ sudo reboot
```

```
$ uname -a
```

```
Linux raspberrypi 5.10.52-v7l+ #1441 SMP Tue Aug 3 18:11:56 BST 2021 armv7l GNU/Linux
```

```
$ ls -l /lib/modules
```

```
drwxr-xr-x 3 root root 4096 Aug 27 02:13 5.10.52+
drwxr-xr-x 3 root root 4096 Aug 27 02:13 5.10.52-v7+
drwxr-xr-x 3 root root 4096 Aug 27 02:13 5.10.52-v7l+
drwxr-xr-x 3 root root 4096 Aug 27 02:13 5.10.52-v8+
```

The headers for the upgraded kernel needed to build the kernel module can be installed from the raspberrypi-kernel-headers package.

```
$ sudo apt install raspberrypi-kernel-headers
```

```
$ ls -l /usr/src
```

```
drwxr-xr-x 23 root root 4096 Aug 24 22:48 linux-headers-5.10.52+
drwxr-xr-x 23 root root 4096 Aug 24 22:48 linux-headers-5.10.52-v7+
drwxr-xr-x 23 root root 4096 Aug 24 22:48 linux-headers-5.10.52-v7l+
```

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Finally, install the remaining required packages.

```
$ sudo apt install vim iperf iperf3
```

NOTE: Kernel Build (Optional)

If the kernel is not up to date, the kernel headers can only be installed manually using the kernel source. The kernel source for Raspberry Pi OS can be found and downloaded at https://github.com/raspberrypi/linux/releases.

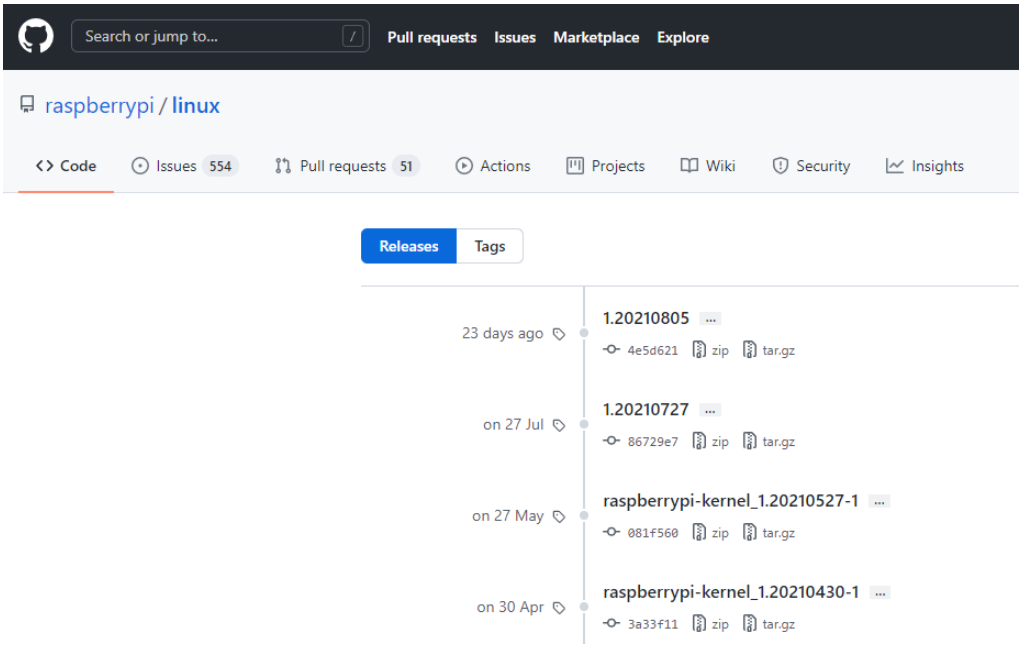


Figure 3.4 Raspberry Pi OS Kernel Source download page

Find and download the kernel source that matches the kernel version of Raspberry Pi OS.

```
$ uname -a
```

```
Linux raspberrypi 5.10.17-v7l+ #1414 SMP Fri Apr 30 13:20:47 BST 2021 armv7l GNU/Linux
```

```
$ cd ~/Download
```

```
$ wget https://github.com/raspberrypi/linux/archive/refs/tags/raspberrypi-kernel_1.20210527-1.tar.gz
```

```
$ tar xzvf raspberrypi-kernel_1.20210527-1.tar.gz
```

```
$ cd linux-raspberrypi-kernel_1.20210527-1/
```

```
$ vim Makefile
```

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```
1 # SPDX-License-Identifier: GPL-2.0
2 VERSION = 5
3 PATCHLEVEL = 10
4 SUBLEVEL = 17
5 EXTRAVERSION =
6 NAME = Kleptomaniac Octopus
```

The downloaded kernel source needs to set up for building external modules. We don't need to change this value.

First, install the required packages.

```
$ sudo apt install git bc bison flex libssl-dev make libcurl4-openssl-dev lzop
```

Prepare the default configuration by running the following commands, depending on Raspberry Pi version.

Raspberry Pi Model 3 B/B+ :

```
$ KERNEL=kernel7
$ make bcm2709_defconfig
```

Raspberry Pi Model 4 B (32-bit) :

```
$ KERNEL=kernel7l
$ make bcm2711_defconfig
```

Use the make target "modules" or "modules_prepare" to prepare the kernel for building external modules.

```
$ make <modules | modules_prepare>
```

NOTE:

"make module_prepare" completes faster than "make modules" but does not generate the Module.symvers file. If the module is built without the Module.symvers file, a warning message is output.

Add a build link to refer to this kernel when building external modules.

```
$ cd /lib/modules/$(uname -r)

$ sudo ln -s ~/Downloads/linux-raspberrypi-kernel_1.20210527-1/build
```

* If a symbolic link is already existed, please delete that in advance and re-create this.

```
$ sudo rm -f build
```

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3.3 Disable Broadcom Wi-Fi and Bluetooth

Broadcom Wi-Fi and Bluetooth can be disabled to use only Newracom Wi-Fi. This is possible with the Device Tree Overlay.

Device Tree Blob Overlay files for disabling Broadcom Wi-Fi and Bluetooth can be found in the /boot/overlays directory.

$ ls -l /boot/overlays/disable-*

```
-rwxr-xr-x 1 root root 1073 Jan  5  2021 /boot/overlays/disable-bt.dtbo
-rwxr-xr-x 1 root root  387 Jan  5  2021 /boot/overlays/disable-wifi.dtbo
```

Device Tree Overlay on the Raspberry Pi is possible by adding the dtoverlay parameter in the /boot/config.txt file.

$ sudo vim /boot/config.txt (Raspberry Pi 4 case)

```
dtoverlay=disable-bt
dtoverlay=disable-wifi
```

$ sudo vim /boot/config.txt (Raspberry Pi 3 case)

```
dtoverlay=pi3-disable-bt
dtoverlay=pi3-disable-wifi
```

After reboot, the wlan0 interface be removed.

$ sudo reboot

$ ifconfig -a

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.123.25  netmask 255.255.255.0  broadcast 192.168.123.255
        inet6 fe80::cf3d:c28b:eb76:f77  prefixlen 64  scopeid 0x20<link>
        ether dc:a6:32:1f:75:a1  txqueuelen 1000  (Ethernet)
        RX packets 75  bytes 8984 (8.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 101  bytes 13911 (13.5 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 4  bytes 444 (444.0 B)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 4  bytes 444 (444.0 B)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

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3.4 Disable User mode SPI device driver

In the NRC7394 Evaluation Kit, the NRC7394 module is connected to SPI Master 0 of Raspberry Pi and is selected by number 0 of the two chip-select pins.

When the User mode SPI device driver is loaded into the kernel, the chip-select pins are in the busy state.

$ ls -l /dev/spidev*

```
crw-rw---- 1 root spi 153, 0 Aug 24 22:23 /dev/spidev0.0
crw-rw---- 1 root spi 153, 1 Aug 24 22:23 /dev/spidev0.1
```

The NRC7394 host driver cannot be loaded into the kernel using the insmod command unless the User mode SPI device driver is removed from the kernel. The Device Tree Blob Overlay file for this is not in the /boot/overlays directory, so it is built from the nrc-rpi.dts file shipped in the NRC7394 Software Package (src/nrc/nrc-rpi.dts, see Section 4.1). The overlay disables the User mode SPI device driver and declares the NRC7394 on SPI Master 0, chip-select 0, with its interrupt (GPIO 5) and reset (GPIO 6) pins. One file serves Raspberry Pi 3, 4 and 5 on every kernel version.

$ cd ~/nrc7394_sw_pkg/src/nrc && cat nrc-rpi.dts

```
/dts-v1/;
/plugin/;
/ {
	compatible = "brcm,bcm2835", "brcm,bcm2708", "brcm,bcm2709", "brcm,bcm2837", "brcm,bcm2711",
"brcm,bcm2712"; /* RPI3, 4, 5 */

	fragment@0 {
		target = <&spidev0>; /* or &spi0 */
		__overlay__ {
			status = "disabled";
		};
	};
	fragment@1 {
		target = <&spi0>;
		__overlay__ {
			status = "okay";
			#address-cells = <1>;
			#size-cells = <0>;
			nrc:nrc-cspi@0 {                        /* Bus Number */
				compatible ="nrc80211";
				reg = <0>;                          /* CS Number */
				interrupt-parent = <&gpio>;
				interrupts = <5 4>;                 /* GPIO Number */ /* IRQ_TYPE_LEVEL_HIGH */
				pinctrl-names = "default";
				pinctrl-0 = <&nrc_irq_pins>;
				spi-max-frequency = <20000000>;     /* Max Freq */
				reset-gpios = <&gpio 6 1>;          /* Reset GPIO Number */ /* GPIO_ACTIVE_LOW */
```

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```
                };
            };
        };
        fragment@2 {
            /*
             * Claim the EIRQ GPIO as an input pin at probe time. On the
             * Pi 5 (RP1) each pad has an Input Enable bit that is OFF by
             * default; without this claim the IRQ logic reads the pin as
             * a constant 0, so a LEVEL_HIGH trigger never fires. Applying
             * this pinctrl state turns Input Enable on. Pull-down keeps the
             * active-high line low while it floats (e.g. during chip reset)
             * to avoid spurious interrupts. Harmless on Pi 3/4 (BCM283x),
             * where brcm,function/brcm,pull use the same encoding.
             */
            target = <&gpio>;
            __overlay__ {
                nrc_irq_pins: nrc_irq_pins {
                    brcm,pins = <5>;        /* GPIO Number */
                    brcm,function = <0>;        /* input */
                    brcm,pull = <1>;        /* pull-down */
                };
            };
        };

        __overrides__ {
            max_speed_hz = <&nrc>,"spi-max-frequency:0";
        };
};
```

$ dtc -@ -I dts -O dtb -o nrc-rpi.dtbo nrc-rpi.dts

$ sudo cp nrc-rpi.dtbo /boot/overlays/

After adding the dtoverlay parameter to the /boot/config.txt file.

$ sudo vim /boot/config.txt (Raspberry Pi 4 case)

```
dtoverlay=disable-bt
dtoverlay=disable-wifi
dtoverlay=nrc-rpi
```

$ sudo vim /boot/config.txt (Raspberry Pi 3 case)

```
dtoverlay=pi3-disable-bt
dtoverlay=pi3-disable-wifi
dtoverlay=nrc-rpi
```

Reboot the system and check the spi devices.

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$ ls -l /dev/spidev*

```
pi@raspberrypi:~ $ ls -la /dev/spi*
crw-rw---- 1 root spi 153, 0 Mar  7 14:17 /dev/spidev0.1
```

NOTE:

When Newracom Wi-Fi operates in host mode, the User mode SPI device driver must be removed from the kernel. However, it does not need to be removed when Newracom Wi-Fi operates in Standalone mode that supports AT Command.

NOTE:

Earlier versions of this guide (Ver 1.2 and before) built the overlay from `newracom.dts`, in two variants: one for kernels before 5.16, where the driver created the SPI device itself, and one for 5.16 and later, where `spi_busnum_to_master()` was removed and the device has to come from the device tree. From Ver 1.3 the guide uses `nrc-rpi.dts`, which is kept with the driver source, for these reasons:

1. **One file for every kernel version.** It always declares the NRC7394 device node, which 5.16 and later require and older kernels also accept.
2. **Raspberry Pi 5 support.** On the RP1 chip each pin's input is disabled by default, so the overlay claims the interrupt pin (GPIO 5) as an input; without it the level-high interrupt never fires. Only `nrc-rpi.dts` carries this setting.
3. **No pin conflict.** `newracom.dts` claims GPIO 5 and 7-11 (the SPI pins included); `nrc-rpi.dts` claims only the interrupt pin. Keeping an older `dtoverlay=newracom` line together with `dtoverlay=nrc-rpi` makes the driver probe fail with "*pin gpio5 already requested*", so replace that line on a board set up with an earlier guide.

On Raspberry Pi OS Bookworm, the overlay directory and `config.txt` are under `/boot/firmware` instead of `/boot`.

3.5 Packages and configurations required for Newracom Wi-Fi

Install hostapd and dnsmasq packages for AP mode.

$ sudo apt install hostapd dnsmasq

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Remove the default configuration file to set the configuration file using the -c option when running wpa_supplicant.

$ cd /etc/wpa_supplicant/

$ sudo mv wpa_supplicant.conf wpa_supplicant.conf.unused

Before loading the NRC7394 host driver into the Kernel using the insmod command, the mac80211 module must be loaded into the Kernel. Add the mac80211 module to always be loaded into the kernel during boot.

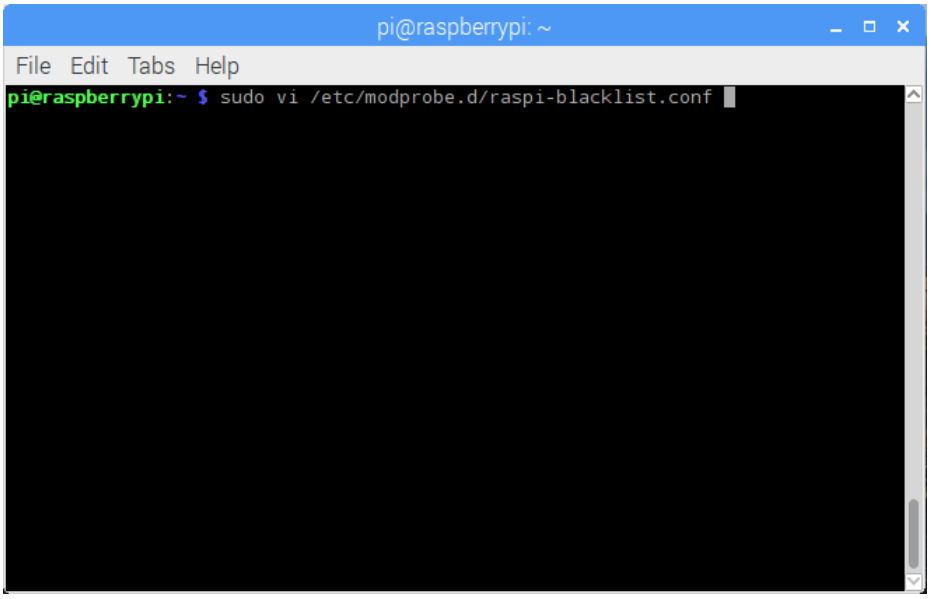
$ sudo vim /etc/modules

```
i2c-dev
mac80211
```

```
1 # /etc/modules: kernel modules to load at boot time.
2 #
3 # This file contains the names of kernel modules that should be loaded
4 # at boot time, one per line. Lines beginning with "#" are ignored.
5
6 i2c-dev
7 mac80211
```

We don’t use existing Broadcom’s WiFi, so please add it into the raspi-blacklist.conf.

$ sudo vim /etc/modprobe.d/raspi-blacklist.conf

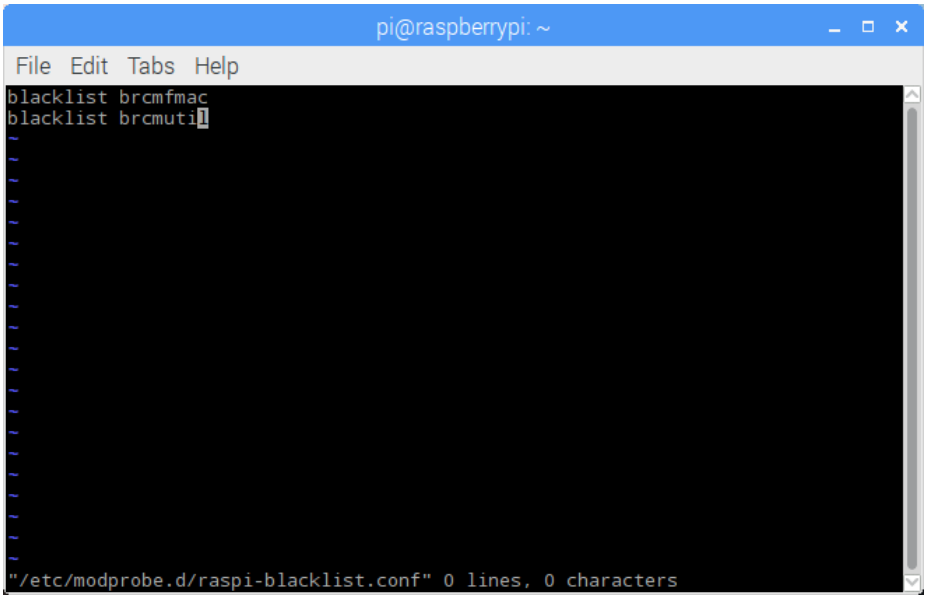


Add blacklist.

```
blacklist brcmfmac
```

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```
blacklist brcmutil
```



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4 Installing NRC7394 Software Package

NRC7394 Software Package contains files for evaluating Newracom Wi-Fi in host mode. After installing it, make sure there are no issues with the configured Raspberry Pi.

4.1 Download and install the package from GitHub repository

$ cd ~

$ git clone https://github.com/newracom/nrc7394_sw_pkg.git

$ cd nrc7394_sw_pkg/evk/sw_pkg

$ chmod +x update.sh

$ ./update.sh

```
Copy new package
apply nrc_pkg
Change mode
Done
```

4.2 Compile and replace the NRC7394 host driver

$ cd ~/ nrc7394_sw_pkg/src/nrc

$ make

```
make[1]: Entering directory '/usr/src/linux-headers-5.10.52-v7l+'
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-mac80211.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-trx.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-init.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-debug.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/hif.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/wim.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-hif-debug.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-hif-uart.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-hif-ssp.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-fw.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-netlink.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-ssp.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-hif-cspi.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/mac80211-ext.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-stats.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-pm.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-dump.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-hif-sdio.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-bd.o
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc-recovery.o
  LD [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc.o
  MODPOST /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/Module.symvers
  CC [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc.mod.o
  LD [M]  /home/pi/nrc7292_sw_pkg/package/host/nrc_driver/source/nrc_driver/nrc/nrc.ko
make[1]: Leaving directory '/usr/src/linux-headers-5.10.52-v7l+'
```

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$ cp -b nrc.ko ~/nrc_pkg/sw/driver

$ ls -l ~/nrc_pkg/sw/driver

```
-rw-r--r-- 1 pi pi 204792 Aug 27 08:39 nrc.ko
-rwxr-xr-x 1 pi pi 213744 Aug 27 08:30 nrc.ko~
```

4.3 Start Newracom Wi-Fi in AP or STA mode

Please refer to the Host Mode User Guide on how to get started.

- Directory : nrc7394_sw_pkg/doc
- File Name : UG-7394-001-EVK User Guide (Host Mode).pdf

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5 Revision History

Revision No	Date	Comments
Ver 1.0	4/5/2023	Initial version
Ver 1.1	8/16/2023	Added additional fixes to dtbo file for supporting kernel version 5.16 and higher.
Ver 1.2	10/12/2023	Change DTS file contents to disable SPI0 only and insert DTS files
Ver 1.3	9/29/2026	Replace the two newracom DTS files with the nrc-rpi.dts overlay kept with the driver source (one file for all kernels, Raspberry Pi 5 interrupt pin, no GPIO conflict)

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